



A SMART MUSHROOM INCUBATION SYSTEM FOR ENHANCED YIELD AND SUSTAINABILITY IN SMALL-SCALE FARMING

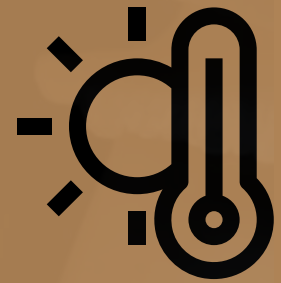
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PP2 Comments and Actions Taken

Comment	Action Taken
Robotic arm did not detect mushroom bag conditions accurately in real-time	Recalibrated the robotic arm and improved focus targeting for accurate detection
Ensure the system works without technical issues at the final stage	Performed testing, debugging, and optimization to ensure stable performance

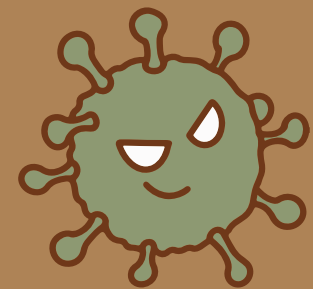
Problem and motivation



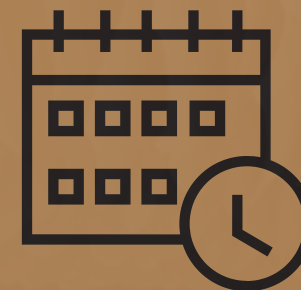
Manual environment
monitoring



Late pest detection



Contamination risks



No fruiting
prediction

Overall Solution over Existing Solutions

Feature	Existing Solutions	Your Solution (AI + IoT System)
Monitoring	Manual or basic sensor monitoring	Real-time IoT-based monitoring
Real-time IoT-based monitoring	Manual inspection, often delayed	Automated real-time pest detection (AI)
Fruiting Prediction	Not available	ML-based fruiting time prediction
Contamination Detection	Late identification	Early detection using CNN models
System Integration	Separate tools / not integrated	Fully integrated smart system
User Interaction	No centralized platform	Mobile/Web dashboard with alerts

Demonstration time



IT22063182 | M L Vidanaarachchi

Predictive Fruiting Module using Environmental Sensing and Machine Learning

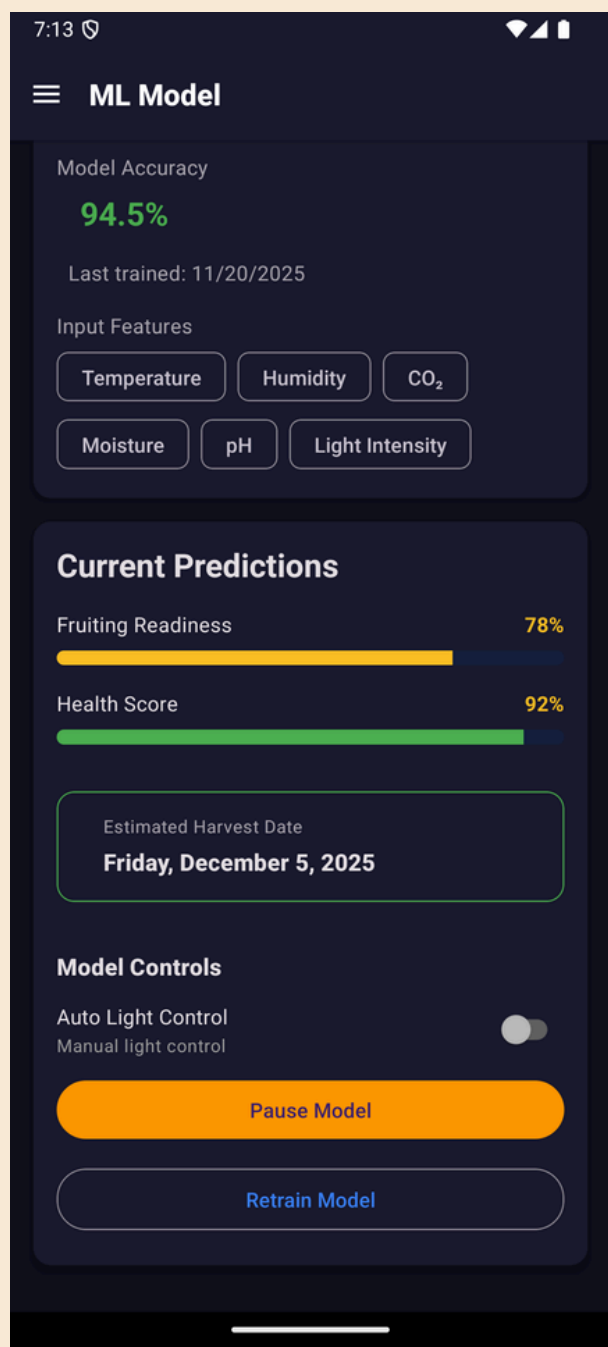
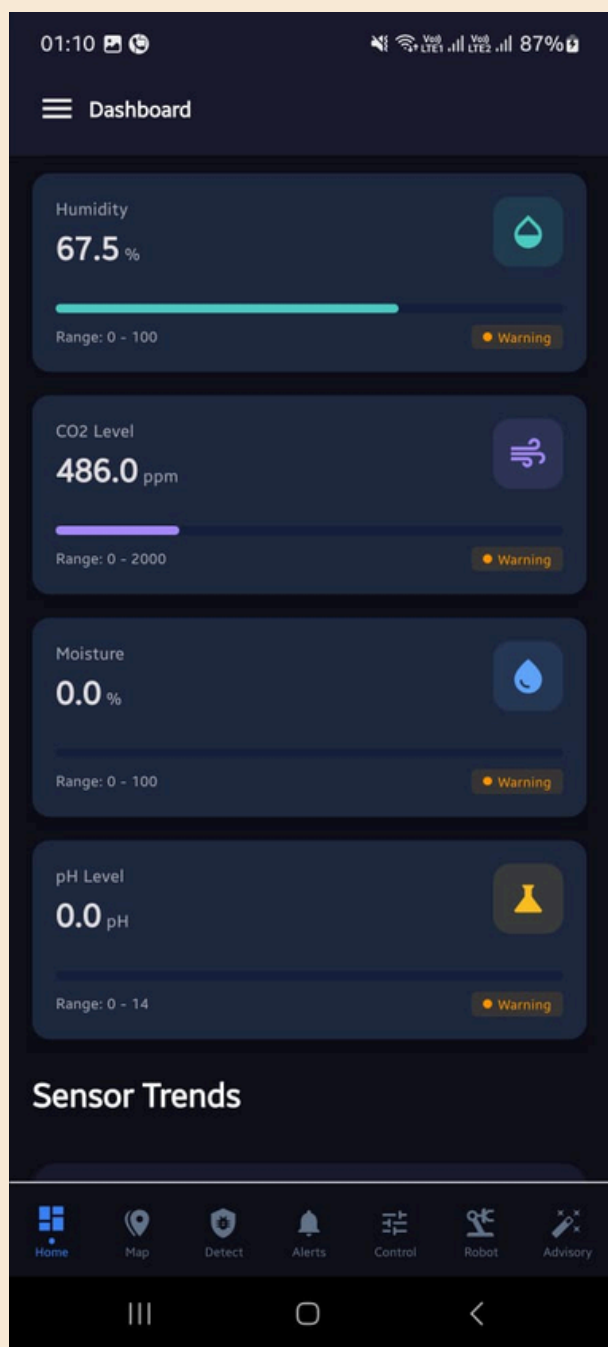
Specialization: Information Technology



Predictive Fruiting Module using Environmental Sensing and Machine Learning

- Monitors real-time environmental conditions in small-scale mushroom farms and predicts the fruiting stage using sensor data (temperature, humidity, etc.).
- Analyzes growth patterns to estimate accurate fruiting timing for better harvest planning.
- Sends timely alerts and updates to farmers to improve decision-making and farm productivity.

Predictive Fruiting Module using Environmental Sensing and Machine Learning



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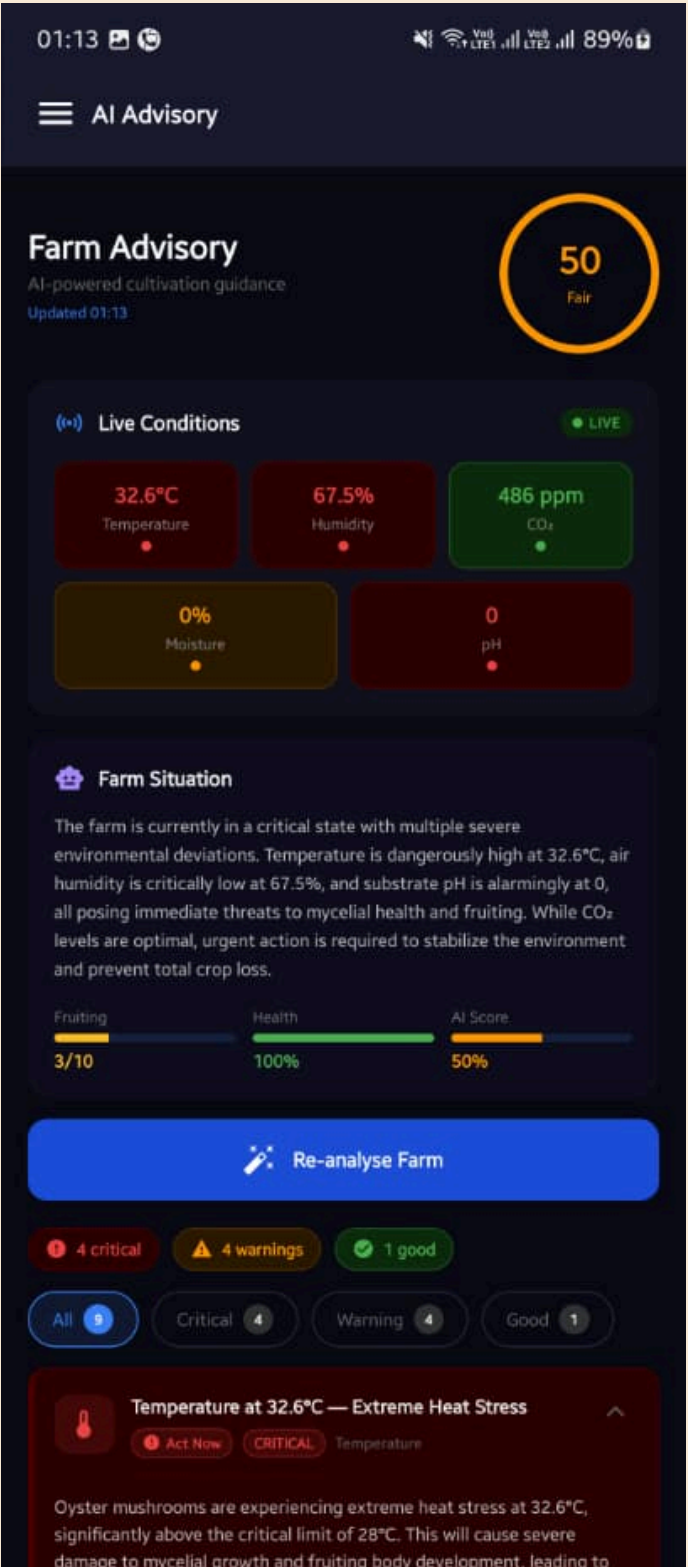
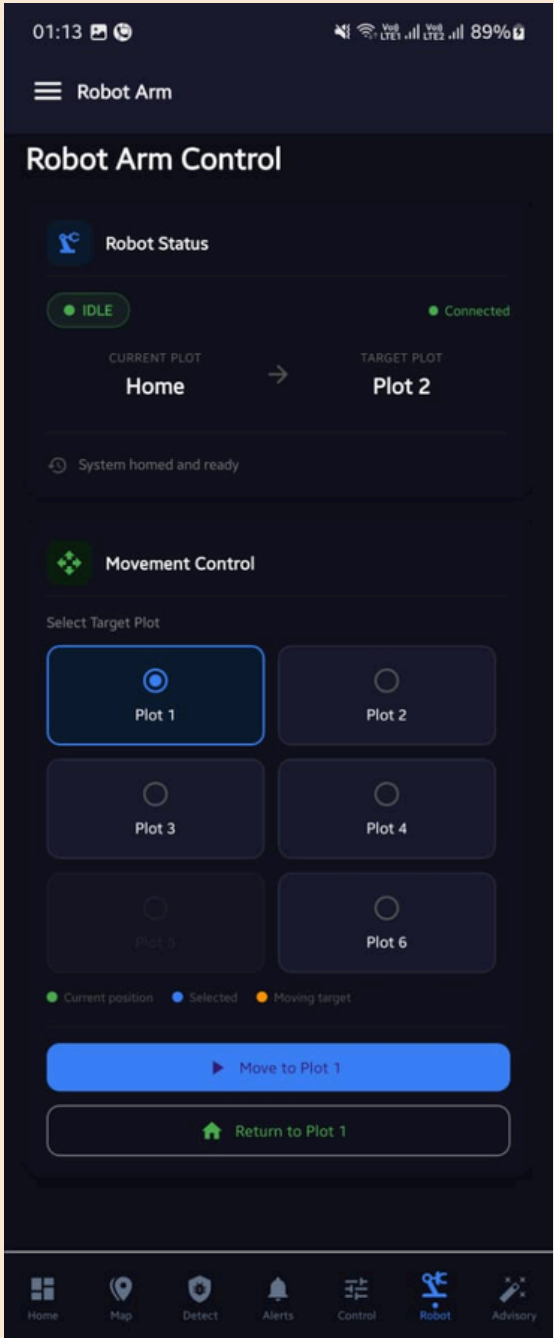
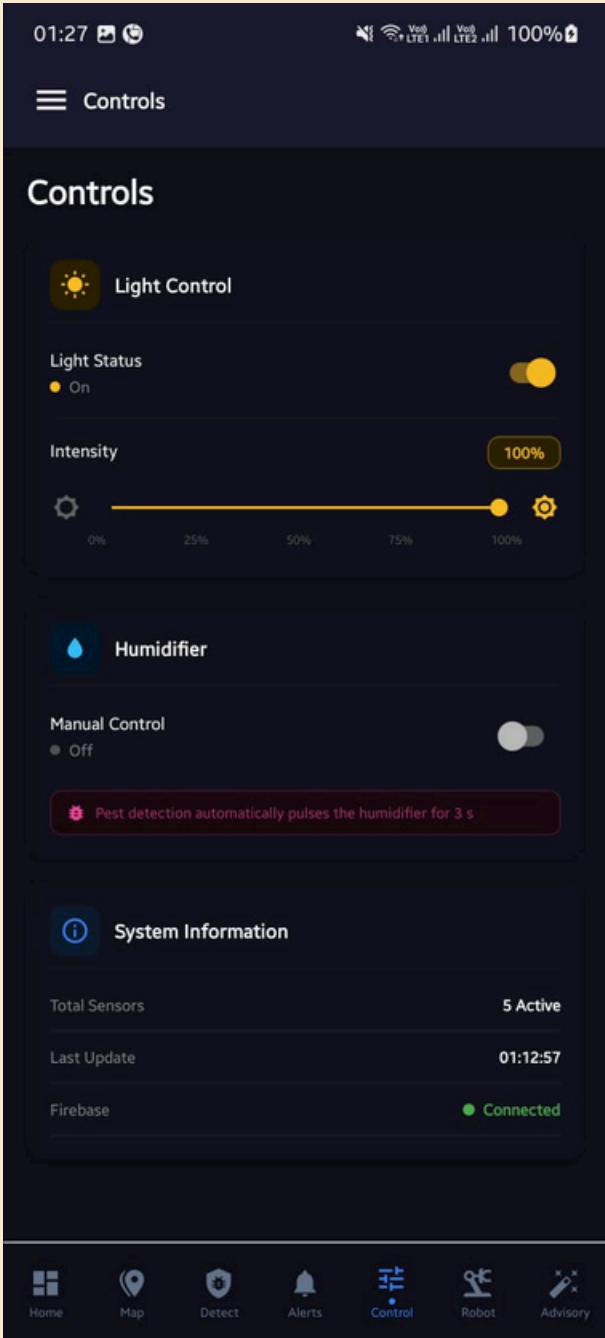
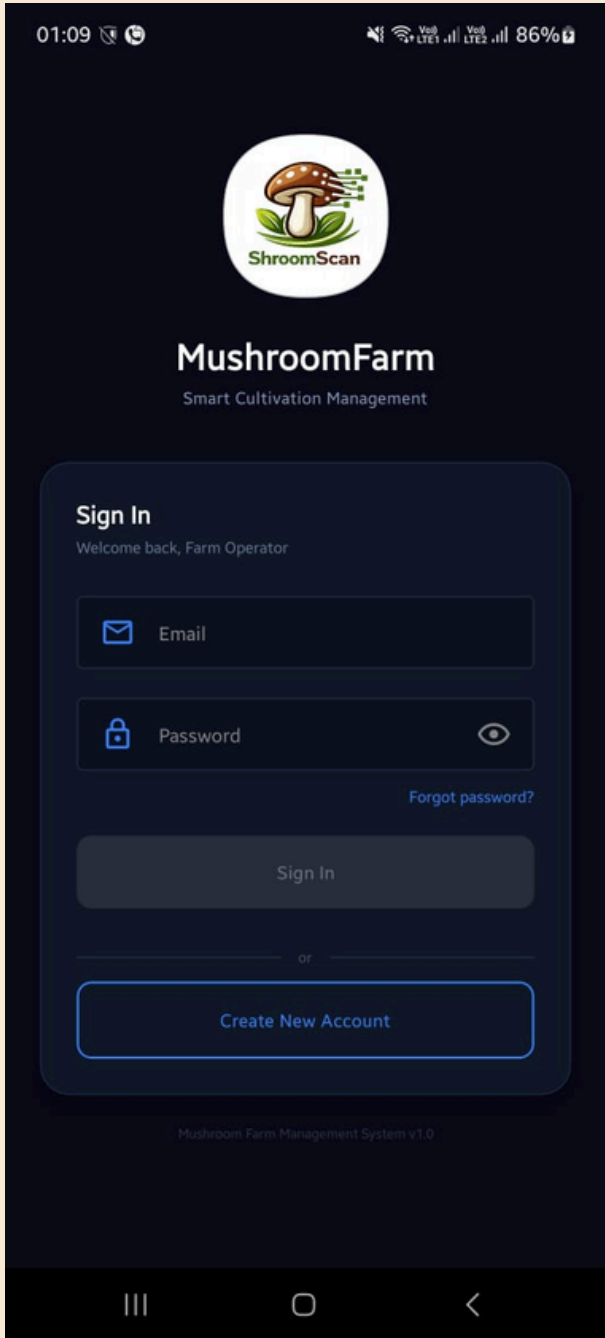
Predictive Advisory System for Moisture and pH Management in Mushroom Cultivation

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Predictive Advisory System for Moisture and pH Management in Mushroom Cultivation

- Monitors moisture levels and pH conditions in mushroom cultivation using real-time sensor data.
- Analyzes environmental conditions to provide predictive insights for optimal growth.
- Gives advisory recommendations to farmers for adjusting moisture and pH to improve yield and quality.

Predictive Advisory System for Moisture and pH Management in Mushroom Cultivation



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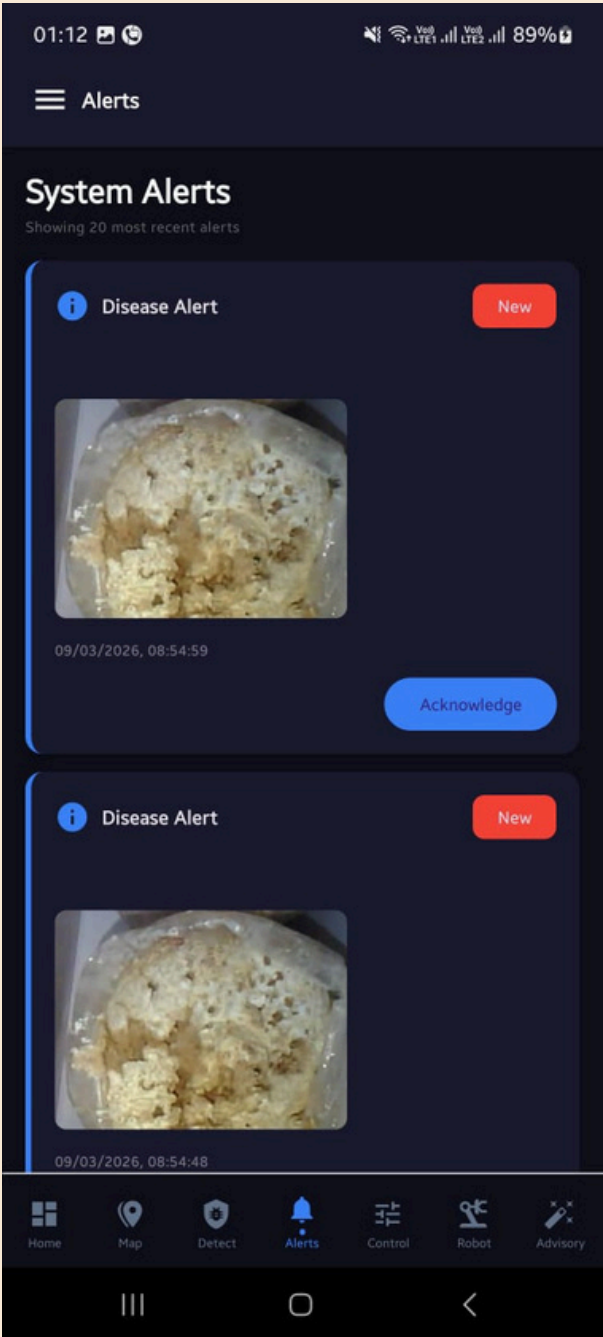
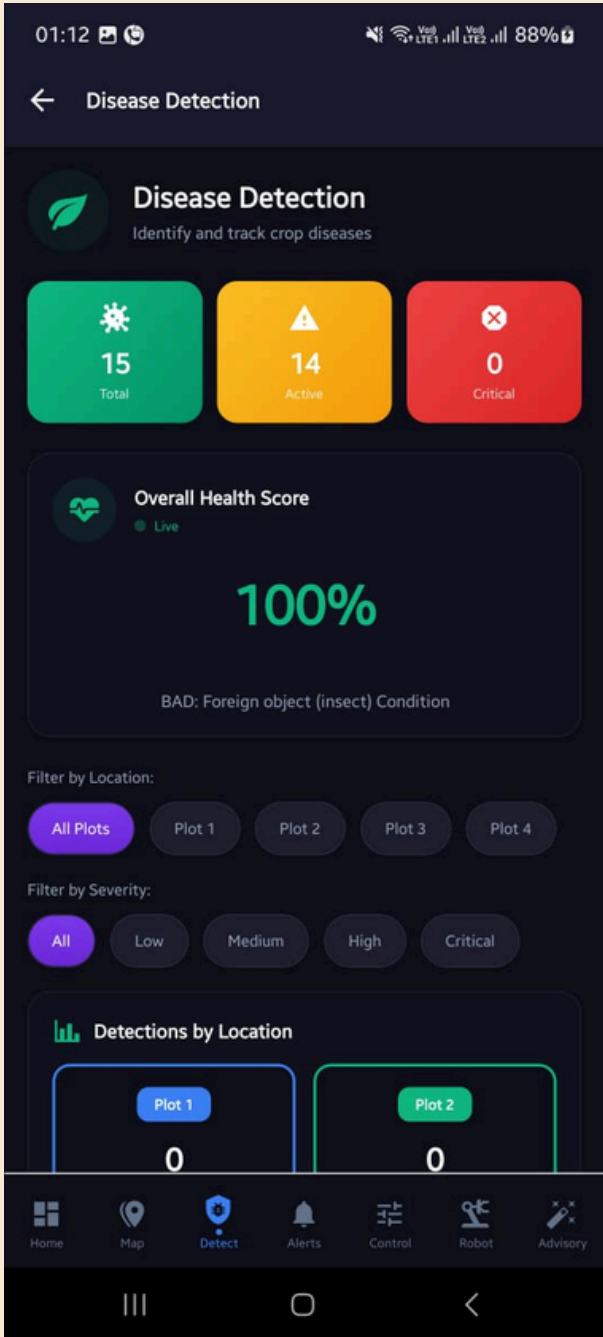
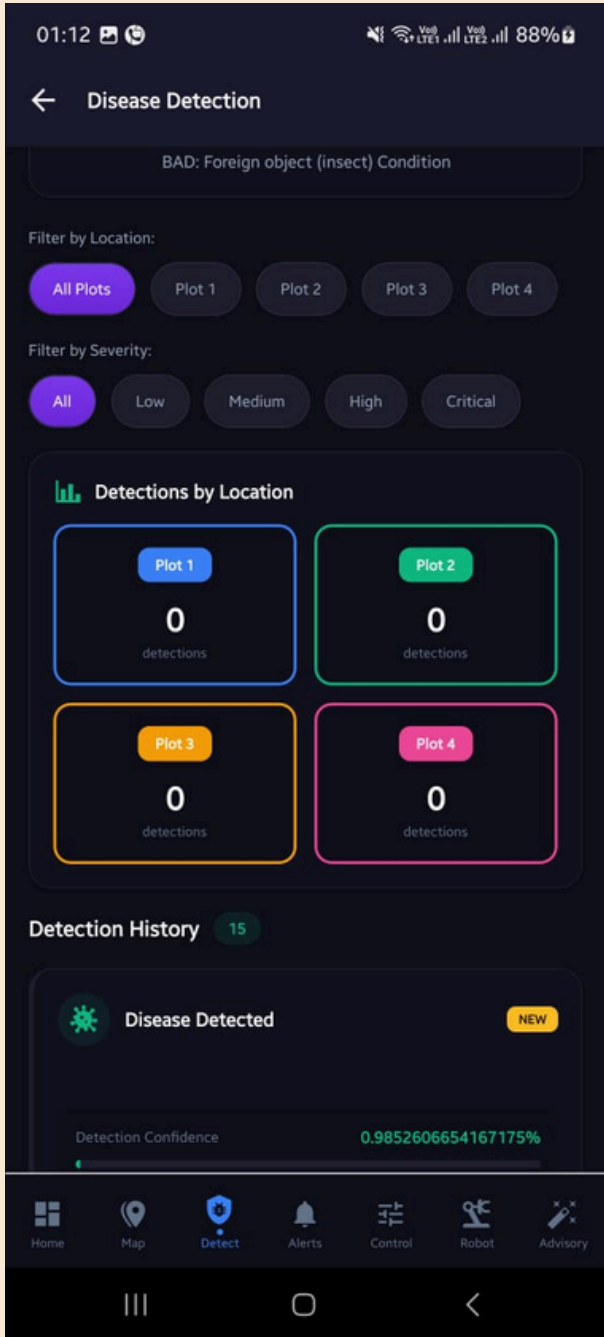
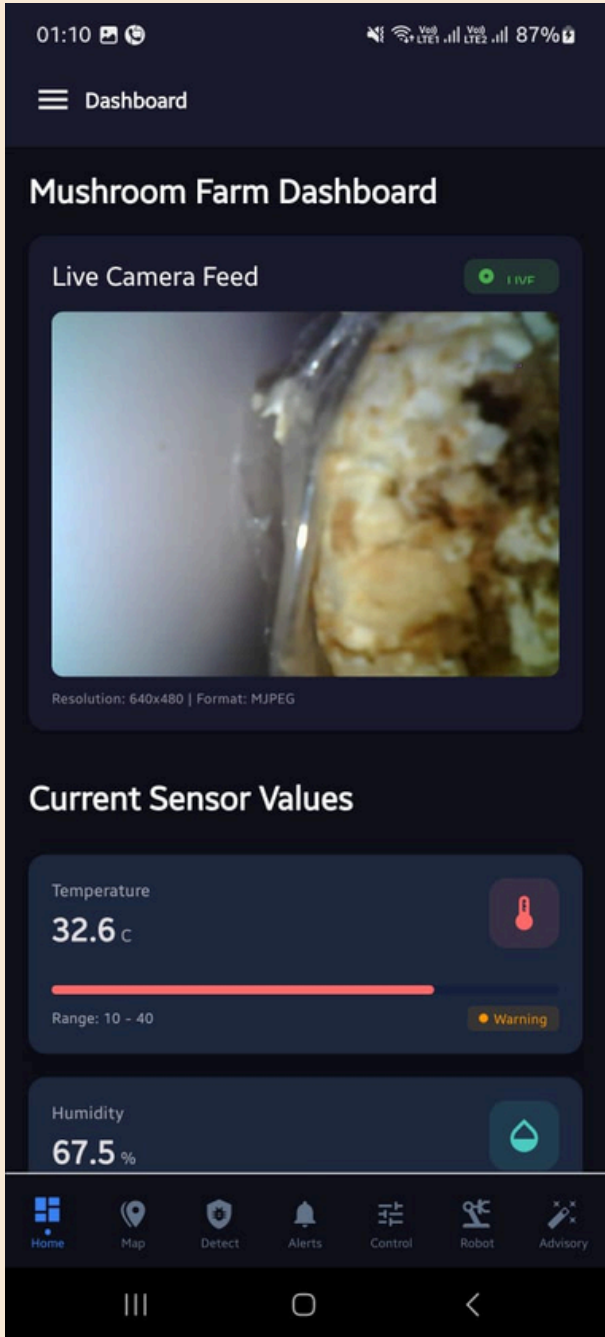
Fungal Contamination Detection Module using Image Processing and Deep Learning

Specialization: Information Technology

Fungal Contamination Detection Module using Image Processing and Deep Learning

- Detects and identifies diseases in mushrooms using a camera-based system with image analysis and monitoring techniques.
- Determines the type of disease and highlights affected areas on the mushroom.
- Sends real-time alerts to farmers with disease details and exact location to enable quick action and reduce crop damage.

Fungal Contamination Detection Module using Image Processing and Deep Learning



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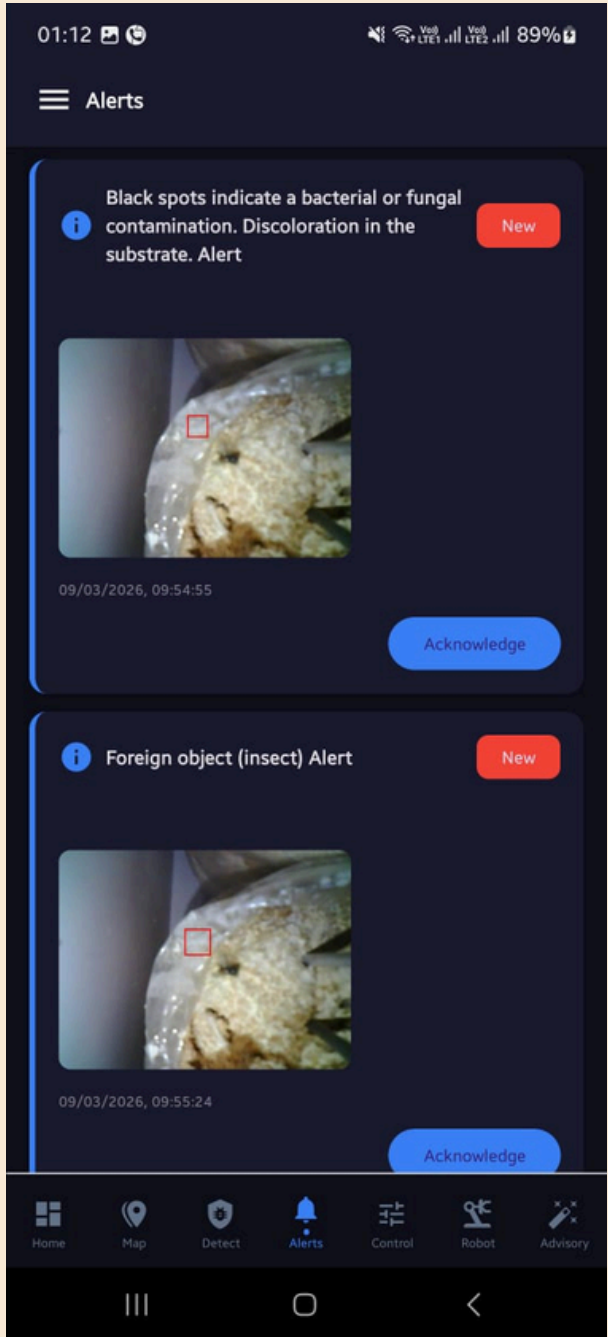
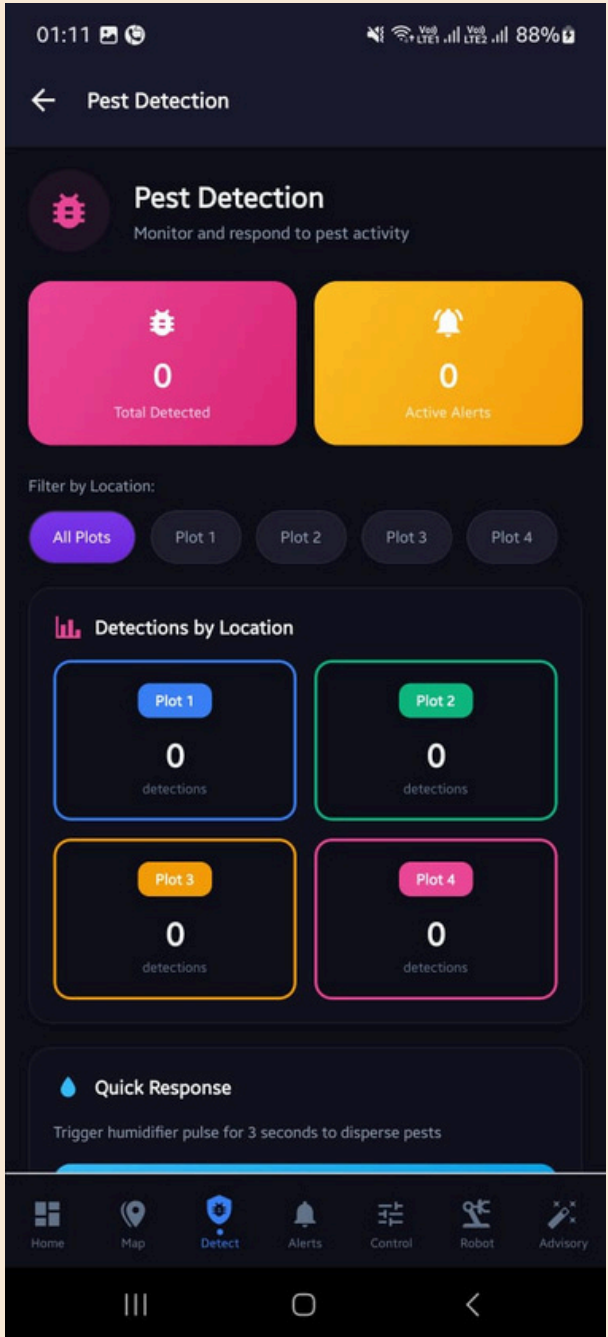
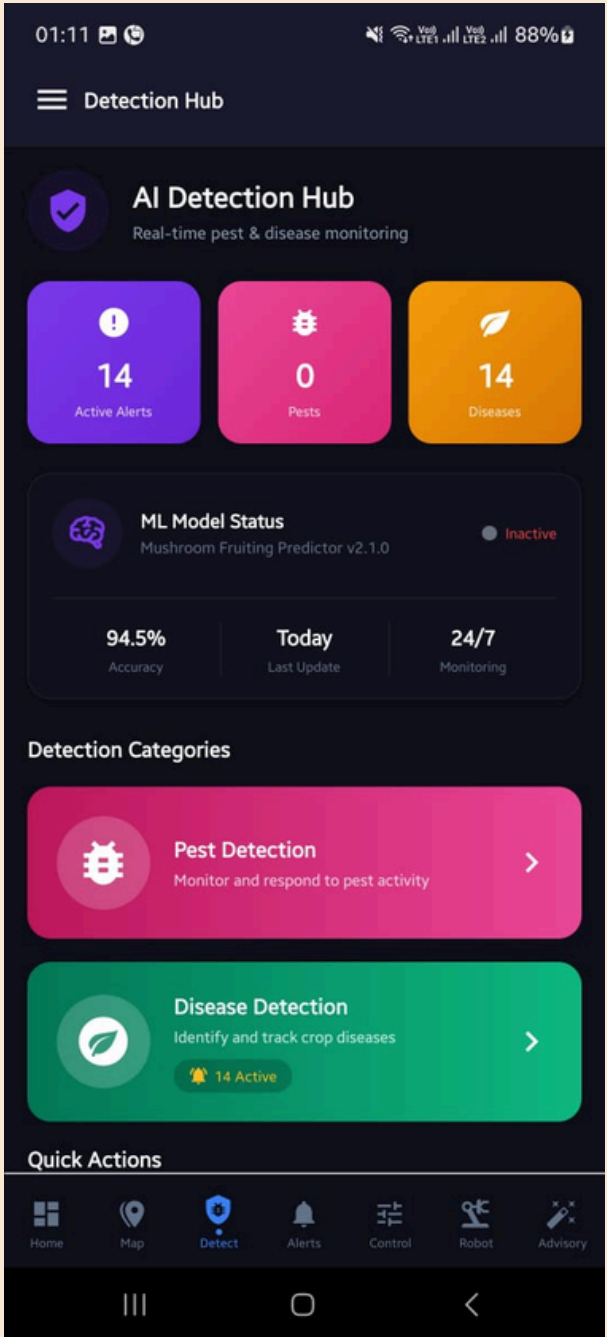
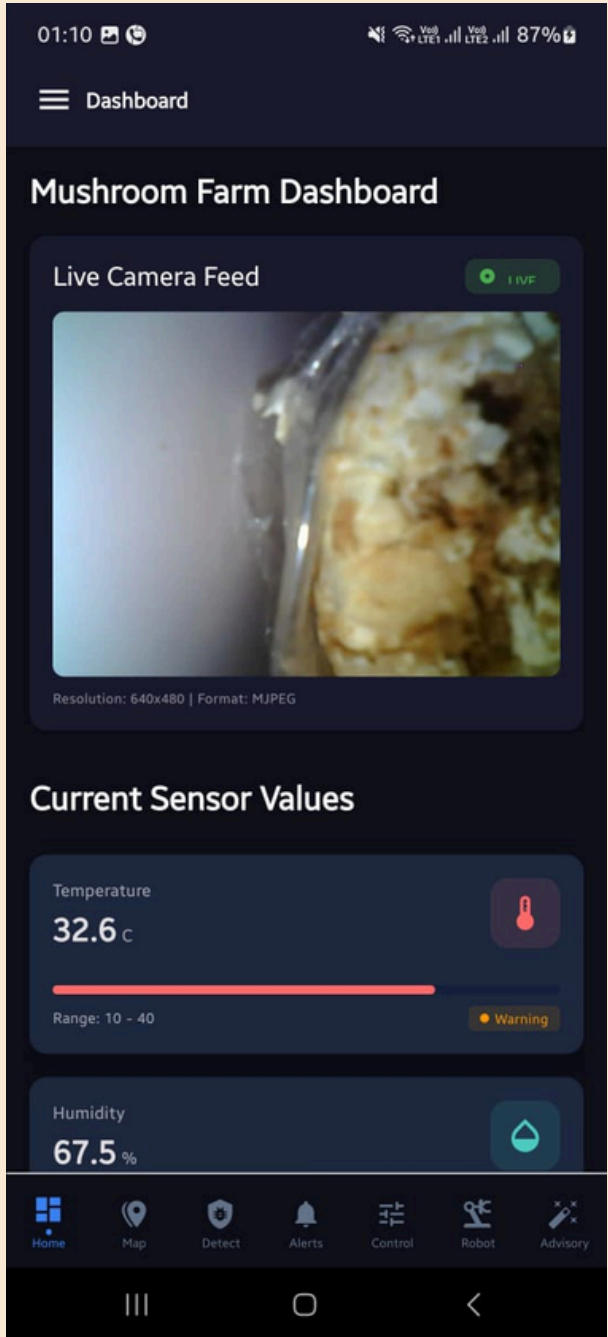
Pest Management Module using Real-Time Detection and Automated Response

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Pest Management Module using Real-Time Detection and Automated Response

- Detects pests in mushrooms using a camera-based image analysis system.
- Identifies pest presence early to prevent spread and crop damage.
- Triggers neem spray recommendations and alerts to help farmers control pests effectively and protect mushroom growth.

Pest Management Module using Real-Time Detection and Automated Response



Limitations of the System and Future Improvements

Limitations of the System

- Requires large training datasets
- Limited real-world testing
- Internet dependency

Future Improvements

- Support for different mushroom varieties
- Offline support
- Solar-powered deployment

Questions?



Thank You!